

Identifying Similar Weather Patterns Using Different p -Wasserstein Distances

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In order to support decision making in airline operations and air traffic management, Wasserstein distances have been proposed for thunderstorm image similarity detection. However, our prior work only investigated the quadratic Wasserstein distance for certain airspace/airport settings. In this paper, we investigate different p -Wasserstein metrics and study their performance using real weather data from the Dallas Fort Worth airspace. In the experiments, we show that the similarity results based on p -Wasserstein distances are more consistent with human perception than existing metrics (SSIM, FSIM, MSE, and KL). Furthermore, we compare the performance of 1-Wasserstein and 2-Wasserstein metrics. Using the real data, we show that our proposed new metric on 1-Wasserstein captures more features of the thunderstorm sequence compared to baseline methods. In addition, we also discuss an example where the 1-Wasserstein and 2-Wasserstein are used to track thunderstorm sequences.

I. Introduction

Severe weather events especially thunderstorms have great impacts on airline operations and air traffic management. A thunderstorm here is defined if the measured Vertically Integrated Liquid (VIL) density is above a certain intensity threshold. In summer air travels, severe thunderstorms are one of the top reasons for large-scale flight delays, cancellations and diversions. Though thunderstorms are highly uncertain, the meteorology community has developed and enhanced thunderstorm predication tools in the past years. However, from the air traffic management and airline flight operation perspectives, thunderstorm prediction is only part of the equation for their decision making [1, 2]. They care more about identifying similar thunderstorm patterns, which will guide them to make operational decisions by referring the current day thunderstorm to similar weather events in the past [2].

More specifically, a potential application scenario is described as follows. The manager of an airline operation center receives a future 2-hour thunderstorm forecast from his/her aviation meteorologist, and they can stitch the past 1-hour thunderstorm pattern (assuming the thunderstorm was formed about 1 hour ago) and the 2-hour forecast together. Now they will have a 3-hour sequence of thunderstorm images. With a tool of identifying similar weather pattern, they can search in the archived weather database to see which days in the past also experienced the similar thunderstorm in the same region like today. More importantly, the airline database also stores what operation decisions were made on those days, such as flight cancellations and diversions. These decisions were recorded together with their operational impacts, such as flight delays, passengers missing connections, disconnected crew sequences, etc. Once we identify the most similar weather days in the past, we can refer to the decisions and decision impacts on those days, and then make an informed data-driven decision for the current operation.

In the literature, there were different ways and metrics to measure the similarity of thunderstorm images (e.g., [3–6]). There are two main categories for these pattern matching methods: the traditional statistical ways using entropy or Markov models, and the ones with machine learning pattern recognition methods. Different methods also adopt different criteria (e.g., entropy-based [6]).

Wasserstein distance (a.k.a. Mallows’ distance, Earth mover’s distance, Monge–Kantorovich distance) is one type of metric to compare probability distributions. It was inspired by the problem of optimal transportation, and is ubiquitous in partial differential equations, fluid mechanics, and probability theory and statistics. In addition to its mathematical importance, Wasserstein distance also provided a successful metric for the comparison of objects, such as image retrieval [7]. In our pioneer work [2], we formulate the thunderstorm image matching problem as an optimal transport problem

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and use Wasserstein distance (only $p = 2$) as a metric to quantify image similarity. Since Wasserstein distance is an optimization problem [2] with high computational complexity, fast ways to solve Wasserstein problem have been proposed in [8, 9]. In our application, we need to make sure that the Wasserstein distance can be computed efficiently and applied to real-world weather events. Furthermore, there is no thorough comparison of Wasserstein distance with other existing metrics in [2].

In this paper, we propose to use 1-Wasserstein, 2-Wasserstein, and mixed Wasserstein distances for thunderstorm image matching and classification. The statistical properties of the three metrics will be investigated. Real radar measurement data is adopted in the simulations [10]. Simulation results show that our new metrics outperform the existing ones in thunderstorm image recognition.

The rest of the paper is organized as follows. In Section II, several metrics and statistical distances to quantify image similarity are introduced. Section III introduces Wasserstein distances and its properties. The real data processing is introduced in Section IV and the comparisons with the existing metrics are performed in Section V. The thorough validation of our proposed Wasserstein distances in large dataset and comparisons with other statistical distances are introduced in Section VI. Concluding remarks and future work are summarized in Section VII.

II. Preliminaries and Related Work

In this section, we list some notations and definitions that will be adopted in the following sections.

A. Notations and Definitions

Notations in this paper are given as follows.

- $\mathbf{x}$: bold lower case letters denote vectors;
- x_n : the n th element of vector $\mathbf{x}$;
- $\mathbb{R}^n$: n -dimensional real space;
- $\|\mathbf{x}\|_p$: L_p norm of the vector $\mathbf{x}$, where $p \geq 0$;
- $\mathbb{E}[x]$: statistical expectation of random variable x ;
- $x \sim \mu$: random variable x follows the distribution μ .

Definition 1. (Monge's Transport Formulation) [11] Consider two probability measures μ and ν on spaces X and Y , respectively. Let $c(x, y)$ be a cost function where $x \in X$ and $y \in Y$ which measures the cost of transporting one unit of mass from x to y . The transport map $\mathbb{M} : X \rightarrow Y$ is defined as

$$\nu(B) = \mu(\mathbb{M}^{-1}(B)), \quad (1)$$

where B is all possible sets which are ν -measurable.

The optimal transport (OT) problem is to find a map from μ to ν with minimum cost $c(X, Y)$. One of the cost functions for OT problem is the Wasserstein distance.

Definition 2. (p -Wasserstein distance): The general p -Wasserstein distance between two probability measures is defined as [12]

$$W_p(\mu, \nu) := \inf_{\mathbf{x} \sim \mu, \mathbf{y} \sim \nu} (\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|_p])^{\frac{1}{p}}, \quad p \geq 1, \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^T$ and $\mathbf{y} \in \mathbb{R}^T$ are M -dimensional random vectors with marginal probability of density functions (PDFs) μ and ν . For discrete distributions and two dimensional objects, the p -Wasserstein distance in (2) can also be written as:

$$W_p(\mu, \nu) := \min_{\{Q_{m,n}\}} \left(\sum_{m=1}^M \sum_{n=1}^N |x_m - y_n|^p Q_{m,n} \right)^{\frac{1}{p}}, \quad p \geq 1, \quad (3)$$

$$\begin{aligned} \text{s.t. } & \sum_{m=1}^M Q_{m,n} = b_n, \quad n = 1, \dots, N, \\ & \sum_{n=1}^N Q_{m,n} = a_m, \quad m = 1, \dots, M, \\ & Q_{m,n} \geq 0, \quad \forall m, n \end{aligned} \quad (4)$$

where $Q_{m,n}$ is the transport plan representing the probability mass transported from x_m to y_n , a_m and b_n are the marginal probability masses.

Definition 3. (Kullback–Leibler (KL) distance for two distributions) The Kullback–Leibler (KL) distance/divergence (a.k.a. relative entropy) is another metric to measure the statistical distance between two distributions:

$$D_{KL}(\mu\|\nu) = \sum_{x \in X} \mu(x) \ln \frac{\mu(x)}{\nu(x)}, \quad (5)$$

where μ and ν are two PDFs.

In this paper, we consider thunderstorm patterns from color radar images. Since our main goal is to match the images with similar thunderstorm patterns, we binarize the images to emphasize the thunderstorms in the image. Therefore, we define the number of thunderstorms in one binary image as follows.

Definition 4. (Number of thunderstorms in an image) The number of thunderstorms in a binarized radar weather image is defined as the number of ones in the binary image.

B. Existing Similarity Metrics

There are existing similarity metrics in image processing. In the following, we introduce three widely adopted image similarity metrics in the literature.

1. Structural Similarity Index Method (SSIM)

One perpetual similarity metric that is widely used in image processing is the Structural Similarity Index Method (SSIM) [13]. This metric evaluates similarity between images based on human perception of structural information. It uses three components to assess similarity: contrast, structure, and luminance. Contrast similarity is measured by the standard deviations of pixel values in each image, indicating how varied or uniform the values are. Structural similarity is evaluated by the covariance which represents differences in the images. Lastly, luminance similarity is measured using the means of the pixel values, reflecting the brightness of the images. These three components are combined into the SSIM formula,

$$SSIM(\mathbf{x}, \mathbf{y}) := \frac{(2\mu_x\mu_y + \epsilon_1)(2\sigma_{xy} + \epsilon_2)}{(\mu_x^2 + \mu_y^2 + \epsilon_1)(\sigma_x^2 + \sigma_y^2 + \epsilon_2)}, \quad (6)$$

where $\mathbf{x}$ is the source image, $\mathbf{y}$ is the target image, μ_x and μ_y are the means of the pixel values, σ_x and σ_y are the standard deviations, σ_{xy} is the covariance between them, and $\epsilon_1, \epsilon_2 > 0$ are small terms to avoid a division by zero.

It is suggested that this function be used locally rather than globally. By applying it across different regions in the image, the values can then be averaged to produce the more robust Mean Structural Similarity Index (MSSIM) below,

$$MSSIM(X, Y) := \frac{1}{M} \sum_{m=1}^M SSIM(x_m, y_m), \quad (7)$$

where x_m and y_m are the m -th regions of the images.

2. Features Similarity Index Method (FSIM)

Different from SSIM to treat all pixels equally, FSIM compares two features to assess image similarity [13].

The first is Phase Congruency (PC), which measures the consistency of phase alignment at different scales within an image by ignoring brightness and contrast. It closely aligns to the way humans identify features because humans prioritize structure (edges and patterns) over intensity (light).

The second feature is Gradient Magnitude (GM), which captures the local rate of change of pixel intensity in an image, detecting sharpness and transitions between regions. This is critical for edge detection. The FSIM score is then calculated as follows,

$$FSIM(\mathbf{x}) := [S_{PC}(\mathbf{x})]^\alpha [S_G(\mathbf{x})]^\beta, \quad (8)$$

where S_{PC} is the harmonic mean of the Phase Congruences, S_{GM} is the harmonic mean of the Gradient Magnitudes, $\mathbf{x}$ is the position matrix including all pixel positions, and α and β are weighting factors that balance PC and GM features.

3. Mean Square Error (MSE)

MSE is a well adopted metric to quantify the difference between two images. Given two images ($\mathbf{x}$ and $\mathbf{y}$), the MSE is defined as

$$MSE(\mathbf{x}, \mathbf{y}) = \frac{1}{T} \sum_{t=1}^T (x_t - y_t)^2, \quad (9)$$

where x_t and y_t are the t -th pixels of the images $\mathbf{x}$ and $\mathbf{y}$. Note that MSE is averaging out the noise effects.

Another often used metric to quantify image similarity is peak signal-to-noise ratio (PSNR). Since in this paper we do not take noise effect into consideration, we will not investigate it.

We will compare these metrics with our proposed Wasserstein distances in the following sections.

C. Existing Statistical Distances For Image Comparisons

Statistical distances (divergences) have been adopted in many applications to compare different probabilities, moment generating functions, statistical models, etc [14]. There are different types of distances, e.g., KL distance, Hellinger distance, chi-square distance, Wasserstein distance [14]. The optimal choice of a distance given a preset goal is challenging. It relies mainly on empirical tests on the datasets. In the following, we mainly review KL distance and Wasserstein distance.

KL distance defined in (5) has a long history and is even well-adopted as a metric in machine learning algorithms these days. It quantifies the difference of two distributions from information theory point of view. It is also the fundamental metric for Akaike Information Criterion (AIC) to estimate model orders and degrees of freedom in a signal/curve [14]. KL distance can also be used to compare the similarity of two images. By calculating the histograms of the two images, we convert the images into PDFs and thus one can calculate KL distance based on (5). However, KL distance has a few statistical limitations. Some local small differences may cause a large change of the KL distance. For two-dimensional images, KL distance does not capture the spatial shift and thus may not match with human perception.

In comparison, Wasserstein distance, also known as the Earth Mover's distance, has caught a lot of attention in the recent literature. It is commonly used to compare histograms or distributions. Beginning in the 1970s, the two-dimensional Wasserstein distance was applied to image comparison [15]. In a string of papers, the computational technique has been discussed and improved upon. For example, the complexity decreased from $O(N^3)$ to $O(N^2)$, where N is the number of pixels in both images [15]. Our paper considers the two-dimensional Wasserstein distance and will compare it with the KL distance.

III. Wasserstein Distances for Similarity of Thunderstorms

In this section, we propose some Wasserstein distances which capture different features of thunderstorms and thus enhance the pattern matching accuracy of the similarity of the thunderstorm images.

A. Wasserstein Distances for Binary Weather Radar Images

Due to the nature of our thunderstorm pattern matching problem, we notice that the number and the location/time of thunderstorms are more important than the real VIL values in each image. To find similar thunderstorm images is to search for two images with similar thunderstorm patterns. Therefore, the color images with VIL numbers are quantified to binary images; i.e., the values of each pixel are quantified to zero or one by setting a threshold of VIL number, where zero means no thunderstorm and one means thunderstorm.

For images with binary values, the Wasserstein distance between two distributions (4) can be simplified as

$$W_p(\mathbf{x}, \mathbf{y}) := \min_{\sigma \in \mathcal{A}_T} \left(\sum_{t=1}^T |x_t - y_{\sigma(t)}|^p Q_t \right)^{\frac{1}{p}}, \quad \forall p \geq 1, \quad (10)$$

where x_m and y_n are the samples from the two distributions, T is the total number of pixels in an image, σ is the permutation of a vector, $\mathcal{A}_T$ is the set contains all possible orders of a vector with length T , and Q_t denotes probability mass for the t th entry of the image.

Among the p -Wasserstein distances, 2-Wasserstein is the most commonly used in the literature (see e.g., [2]). Here is the 2-Wasserstein distance definition:

$$W_2(\mathbf{x}, \mathbf{y}) := \min_{\sigma \in \mathcal{A}_T} \left(\sum_{t=1}^T |x_t - y_{\sigma(t)}|^2 Q_t \right)^{\frac{1}{2}}, \quad (11)$$

Note that although binarization simplified the problem in (4), it still has high complexity since the set of $\mathcal{A}_T$ is large. Some faster Wasserstein solutions were proposed [9, 15]. Thanks to binarization, the Wasserstein distance is linked to L_p norms.

B. 1-Wasserstein and Its Properties

For binary images, the 1-Wasserstein distance from (10) becomes*:

$$W_1(\mathbf{x}, \mathbf{y}) := \min_{\sigma \in \mathcal{A}_T} \left(\sum_{t=1}^T |x_t - y_{\sigma(t)}| Q_t \right). \quad (12)$$

When the image is binarized using the VIL intensity threshold (see Sec. IV for detailed explanation), it becomes sparse, i.e., only a few non-zero entries. However, the well adopted W_2 (e.g., in [2]) tends to ignore the small terms and focus on the large ones. It is well-known that L_1 norm takes advantage of the sparsity and produces a solution with a smaller number of non-zero entries (see e.g., [16] and references therein). For our applications, the number of nonzero pixels after binarization in one image is a critical parameter for similarity (see Section IV for detailed explanations). This motivates us to consider 1-Wasserstein distance, W_1 for similarity detection. In addition, by exploiting the sparsity, the complexity of finding Wasserstein distance is reduced to between $O(N^2)$ and $O(N)$ [15].

In the following, we show some useful properties of W_1 .

Property 1. For two random vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^T$, given a vector $\mathbf{w} \in \mathbb{R}^T$,

$$W_p(\mathbf{x} + \mathbf{w}, \mathbf{y} + \mathbf{w}) = W_p(\mathbf{x}, \mathbf{y}). \quad (13)$$

Using the definition of Wasserstein distance in (2), it is not hard to prove this property. Property 1 shows that when two images are shifted together, the Wasserstein distance is not changed.

Property 2. Given two random vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^T$, Wasserstein distance W_1 is less than or equal to Wasserstein distance W_2 .

Proof: Given the 2-norm and 1-norm definitions and using Jensen's inequality, we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{x} - \mathbf{y}\|_2^2] &= \mathbb{E} \left[\sum_{t=1}^T (x_t - y_t)^2 \right] \\ &= \sum_{t=1}^T \mathbb{E} [(x_t - y_t)^2] \\ &\geq \sum_{t=1}^T (\mathbb{E} [|x_t - y_t|])^2 \\ &\geq \left(\mathbb{E} \left[\sum_{t=1}^T |x_t - y_t| \right] \right)^2 = (\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|_1])^2. \end{aligned}$$

This shows that given two images, W_1 is less than W_2 . Our experiments in Sections V and VI also confirm this property.

Property 3. Wasserstein distance W_1 takes into account the number of thunderstorms (defined in Definition 4) in the binary image.

*Note that the formulation in (12) can be extended to different weights [16].

It is known that L_1 norm has a direct relationship with sparsity [16]. Due to the special shape of the L_1 norm constraint (diamond shape with the “sharp corners”) for the parameter space, when searching for the solutions, it tends to give sparse solutions. Binary thunderstorm images generate sparsity of thunderstorms. W_1 keeps this feature when comparing one with another. Property 3 shows that W_1 captures a great feature of the thunderstorm pattern. This will be validated by the simulations.

Property 4. *Wasserstein distance W_p of binary images is almost surely consistent with the one W'_p of the original images; i.e., if $W_{p1} < W_{p2}$, then $W'_{p1} < W'_{p2}$.*

Based on Property 1, we notice that Wasserstein distances are shift-invariant. Therefore, when we quantize the images to binary, Wasserstein distances keep the order. Notice that the thunderstorm pattern is sparse and the changes (w term in (13)) from color images to binary images are almost the same. This justifies that if we select the most similar one from binary images, it is also the most similar one in the corresponding original image.

There are other statistical properties [12] of W_1 which may not show direct impact to our current thunderstorm pattern selection, but they are important for our future work on for machine learning based classification methods.

C. Modified 2-Wasserstein with W_1 Regularization

It is shown that W_2 has the advantage in the mean squared sense. To take into account, we propose a modified Wasserstein distance by adding W_1 regularization on the vectors:

$$W_{mix}(\mu, \nu) := \min_{\sigma \in \mathcal{A}_T} \left(\sum_{t=1}^T (|x_t - y_{\sigma(t)}|^2 + \lambda |x_t - y_{\sigma(t)}|) Q_t \right), \quad (14)$$

where λ is chosen to trade-off the squared error and absolute error, and σ is the permutation operation of a vector. Here λ controls the tradeoff between W_1 and W_2 and (14) subsumes W_1 and W_2 as special cases.

IV. Thunderstorm Data Processing

In this paper, we adopt the real weather radar data from [10]. The Multi-Radar/Multi-Sensor System (MRMS) data have two-minute resolution and were provided by NOAA/National Severe Storms Laboratory in GRIB2 format [10]. The weather radar measures the Vertically Integrated Liquid (VIL), which is the total mass of precipitation in the clouds. The VIL density is defined as VIL normalized by the echo top (the height of the cloud to return the radar signal) [17]. It is shown that VIL density is highly correlated with the possibility of thunderstorms [18]. Thus, VIL density is widely adopted as the indicators for thunderstorms [3].

Notice that when VIL density is higher than a certain threshold, thunderstorm occurs with a high probability (almost surely) [18]. Based on this feature, when we compare two thunderstorm images, the differences between the values of the VIL densities are not as important once the VIL densities are over a certain threshold. What is more important is whether the thunderstorm happens or not. Therefore, one may consider to quantify the real valued images into binary ones with 0 meaning no thunderstorm and 1 meaning thunderstorm. In addition, when we consider real values of VIL density, the general Wasserstein distance calculation is computational complex. Thanks to the properties of Wasserstein distances shown in Section III.B, comparing on the binary images is equivalent to the original images.

In detail, for each VIL density image, we just binarize it based on the threshold $3kg \cdot m^{-3}$ based on the results in [18]. The more general case will be considered for future work. In Property 4, we show that the order of the values of Wasserstein distances in the color images is almost surely consistent with the ones in binary images.

V. Comparisons with Existing Similarity Methods

In this section, we use some real radar thunderstorm images and compare the image similarity results from different metrics we have mentioned in Sec. II.

We choose three images from the data set (see Fig. 1 (a) as Source image; (b) as Target Image 1, and (c) as Target Image 2). Six metrics are applied to justify the similarities between Source Image and Target Image 1, and Source Image and Target Image 2: SSIM in (6), FSIM in (8), MSE in (9), 1-Wasserstein distance W_1 in (12), 2-Wasserstein distance W_2 in (11), and KL distance in (5) criteria. Note that among these metrics, higher similarity means larger SSIM and FSIM, but smaller MSE, W_1 , W_2 , and KL distance.

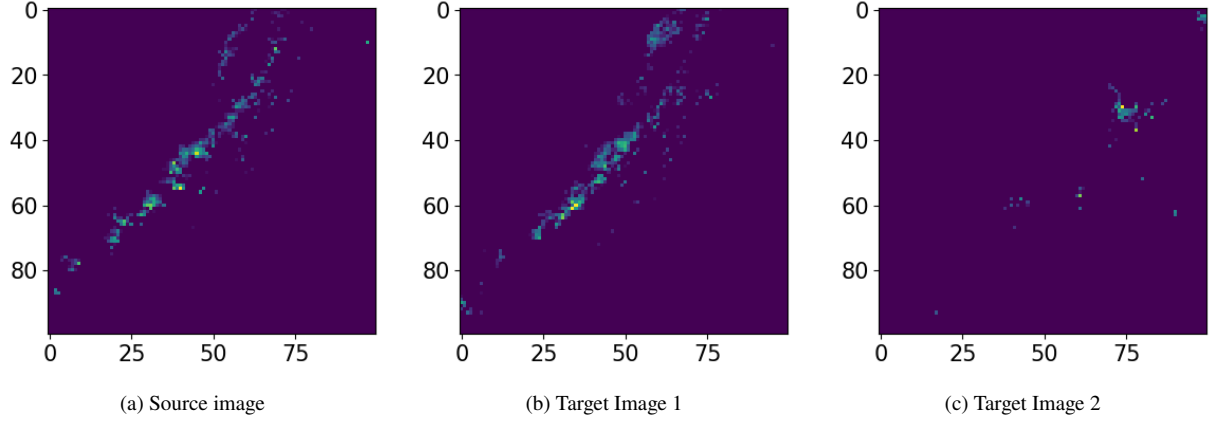


Figure 1 Images to compare the metrics of similarity

Judging by human perception on the three images in Fig. 1, it is obvious that Target Image 1 in Fig. 1b is more similar to the Source Image in Fig. 1a than Target Image 2 in Fig. 1c.

The similarities of Source vs. Target Image 1 and Source vs. Target Image 2 results from the six metrics are shown in Table 1 where boldface numbers mean higher similarity based on the metrics.

Metrics	SSIM	FSIM	MSE	KL	1-Wasserstein W_1	2-Wasserstein W_2
Source vs. Target 1	0.9696	0.9394	136.12	91.76	5.34	25.27
Source vs. Target 2	0.9764	0.9286	65.81	109.54	32.59	816.43

Table 1 Comparisons of SSIM, FSIM, MSE, KL distance, 1-Wasserstein, and 2-Wasserstein

From the table, we have the following observations.

- SSIM result shows Source Image is more similar to Target Image 2 which is against the human perception. One limitation of SSIM is that it treats every pixel with equal importance, despite the fact that humans visually focus on certain regions of an image over others. For instance, low-level features such as edges and zero crossings of an image.
- FSIM captures more features and thus the result is consistent with human perception by confirming Source Image is more similar to Target Image 1. However, the FSIM numbers are pretty close and thus small changes in the image may cause different results.
- MSE is consistent with SSIM which considers Source to be more similar to Target Image 2. This is not surprising since MSE is a more effective metric to handle noise, but not shifting or other image feature changes. Therefore, compared with Source Image, Target Image 2 has smaller errors in terms of values in pixels.
- KL distance gives the result that Source Image is closer to Target Image 1. As we described in Sec. V, KL distance compares two images by converting the values of the pixels into histograms. Given the sparsity of the images in Fig. 1, apparently no thunderstorm pixels (the dark blue regions) dominate the image. At the same time, similar to MSE, KL distance also does not take into account the location shifts in the images. Thus, a small change may cause a large KL distance. This is confirmed by the values of the KL distances in Table 1. Although Target Image 1 is much closer to Source Image, the KL value is not that far from the KL value of the Target Image 2. The detailed mathematical comparisons between KL and Wasserstein can be found in [19] and references therein.
- Wasserstein distances W_1 and W_2 both show that Source Image is more similar to Target Image 1. Thanks to the optimal transport problem formulation, Wasserstein distances incorporate the geometry of the space where the random variables are generated. Wasserstein distance W_p quantifies the minimum distance by mapping one image to another one. This property implies that Wasserstein distances can capture the human perception of whether the two images are similar or not (more mathematical proofs can be found in [12]).

VI. Comparisons with Different p -Wasserstein Distances

In Section V, we show the advantages of Wasserstein distances over other metrics by performing closer to human perception. One natural question is: do all Wasserstein distances have the same performance on similarity? In Section III, we illustrate that different Wasserstein distances capture different features of the images. In this section, we compare the performance of 1-Wasserstein distance, 2-Wasserstein distance and mixed Wasserstein distance.

The dataset for the simulation here is the VIL density images during 2022 [10]. We choose the image on 2022-03-07 04:50 as the source image (shown in Fig. 2 (a)). The reference dataset contains the weather images from June 2022 to July 2022 (total 495 images). By computing the Wasserstein distances between each reference image in the dataset and the source image, we choose the one with the smallest Wasserstein distance in (10) as the most similar one.

A. 1-Wasserstein W_1 vs 2-Wasserstein W_2

Among the images in the reference dataset, Fig. 2 (b) is the one with the smallest W_1 in (12) and Fig. 2 (c) is the one with the smallest W_2 in (11) (The calculation of W_2 is the same as the one in [2]).

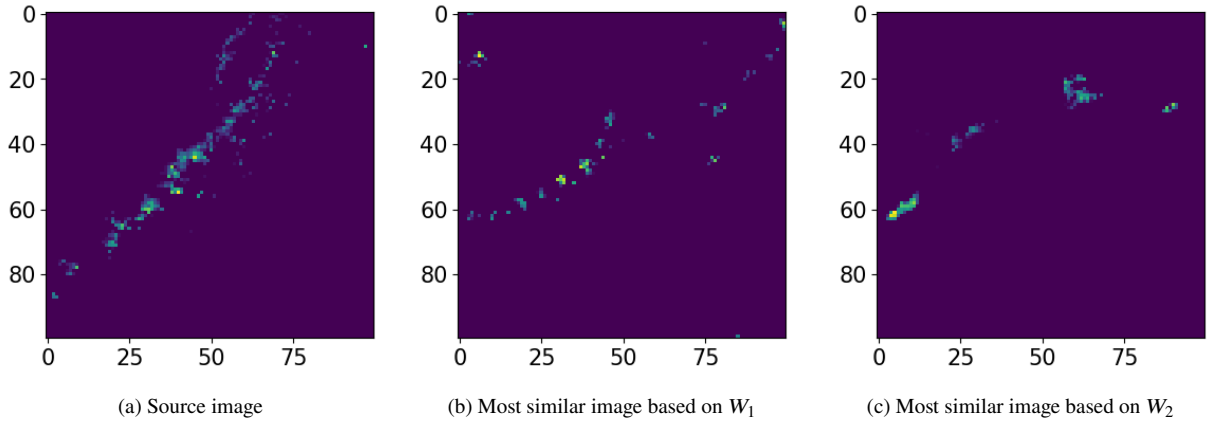


Figure 2 Applying W_1 and W_2 distances to identify the most similar images given a source image.

From this simulation, we have the following observations.

- Consistent with our expectation in Section III, both Wasserstein distance metrics pick up the images which have certain similarity with the source image, although different Wasserstein distances pick up different images as the most similar ones;
- It is shown from the images that compared with image in Fig. 2 (a), our proposed W_1 metric captures the sparsity of the image and thus chooses the one (Fig. 2 (b)) with similar pattern based on human perception. However, the image chosen by W_2 metric in Fig. 2 (c) only captures the main trend of the Source Image. Thus, W_1 outperforms W_2 .
- Although both W_1 and W_2 are calculated on binary images, the original images also show the similarities. This confirms our analysis in Section III.A. Binarization of the images not only reduces the computational cost, but also reserves the order of the Wasserstein distances.

Furthermore, we plot the optimal transport (OT) maps between the binarized source image and the two selected most similar images in Fig. 3. In Fig. 3 (left), the blue dots are the thunderstorms from the binary source image (the original image is given Fig. 2 (a)) and the red dots are the thunderstorms happening in the most similar image selected by W_1 (the original image is given Fig. 2 (b)). In Fig. 3 (right), the blue dots are the thunderstorms from the binary source image (the original image is given Fig. 2 (a)) and the red dots are the thunderstorms happening in the most similar image selected by W_2 (the original image is given Fig. 2 (c)).

We have several observations from Fig. 3.

- Compared with the original radar images in Fig. 2, the binary images show the thunderstorms and keep this main feature.
- Compared with W_2 , the most similar image based on W_1 keeps some long distance non-zero pixels (the upper left corner and lower right corner pixels). However, the image selected by W_2 does not have these pixels because 2-Wasserstein distance W_2 puts high penalty on the long distances and thus avoid these images.

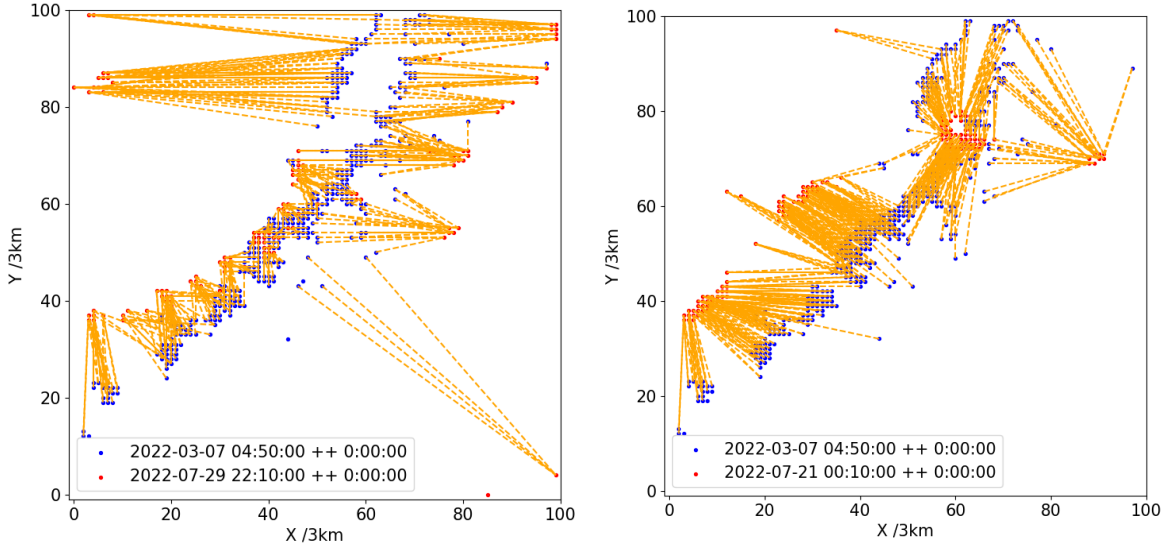


Figure 3 The optimal transport (OT) maps with different Wasserstein distance metrics. (left) OT map to the most similar image based on W_1 ; (right) OT map to the most similar image based on W_2

- Compared with W_1 , the image with lower W_2 has almost all non-zero pixels close to the locations of the non-zero pixels of the source image.
- The most similar image selected by W_1 has the non-zero pixels spread along the pattern of source image, while the non-zero pixels in the selected image by W_2 are more grouped.

B. Mixed Wasserstein distances

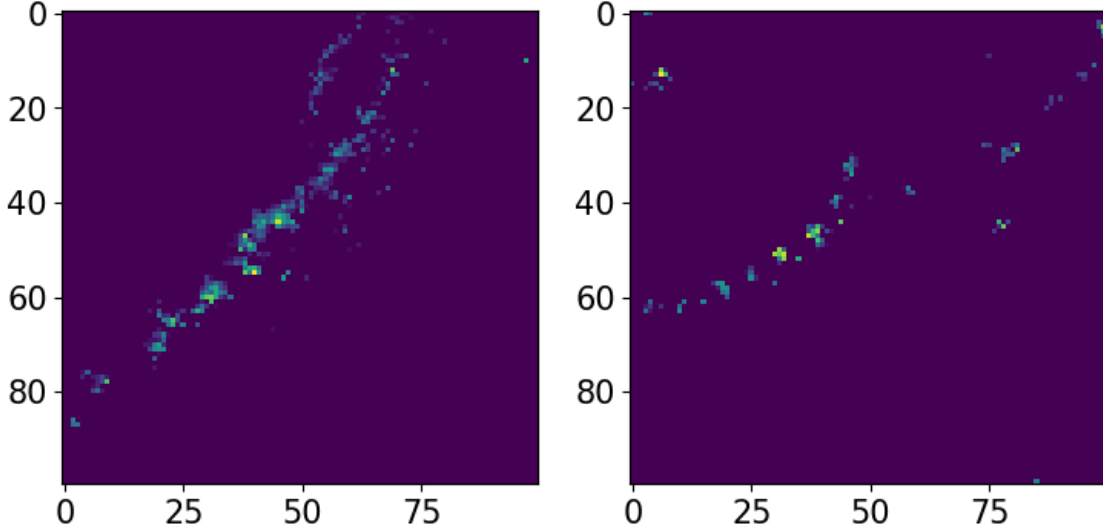


Figure 4 The optimal transport map with mixed Wasserstein distance metric

Using the same dataset with 495 images, we also test the mixed Wasserstein distance in (14). When we set $\lambda = 0.5$ (less weight to W_1), the image with the minimum mixed Wasserstein distance is the same as the one with W_1 , shown in the following. This also shows that 1-Wasserstein distance plays an important role in the pattern matching with sparsity. Further study of this mixed metric is out of the scope of this paper.

C. Comparisons of p -Wasserstein Distances on Image Sequences

Thunderstorm sequence tracking is another important topic for weather analysis. Although in this paper, we focus on thunderstorm image (or snapshot) similarity metrics, we want to investigate whether the ones selected by different Wasserstein distances can predict the trend of thunderstorms by comparing their following sequences.

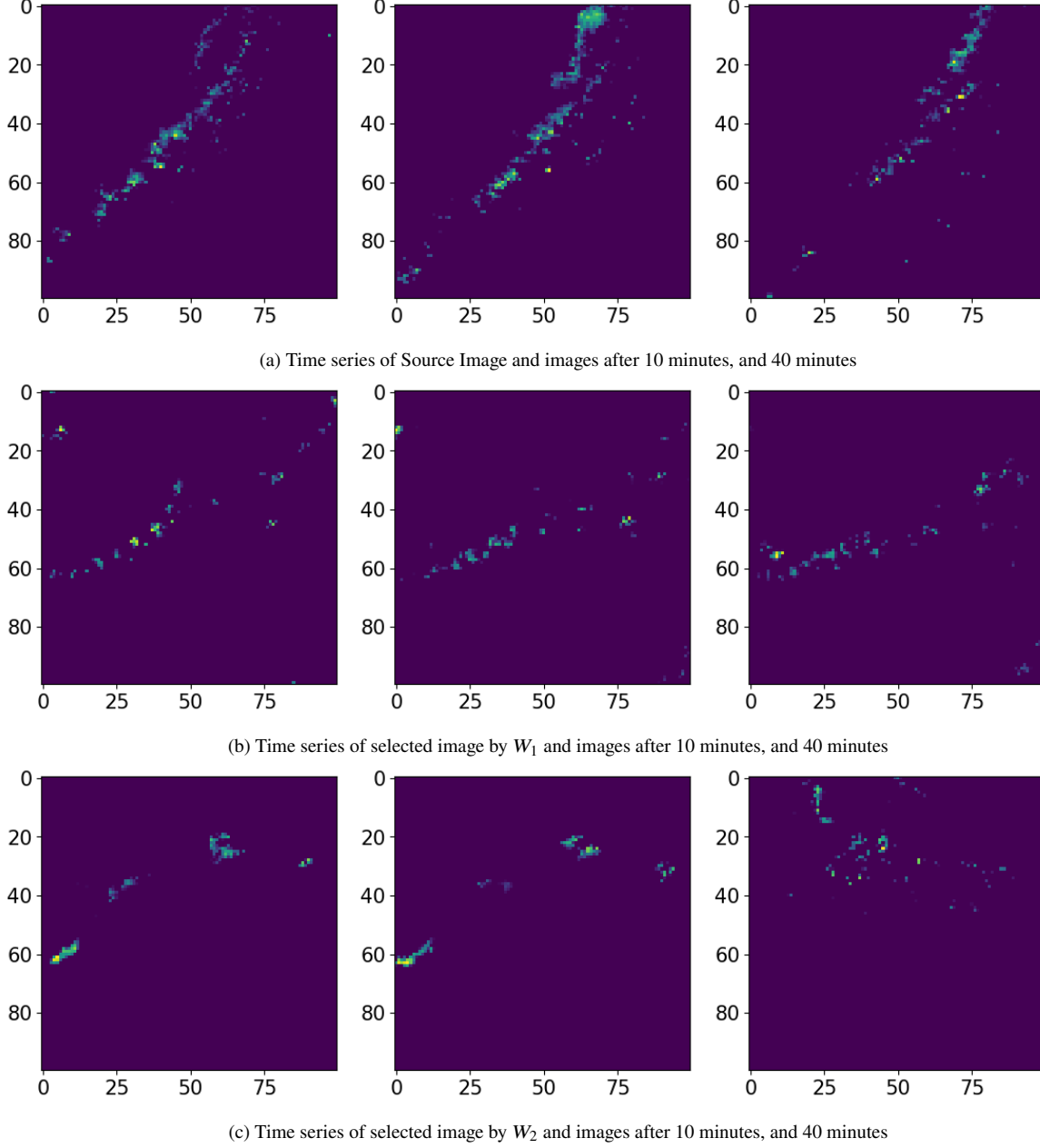


Figure 5 Three thunderstorm image sequences

In Fig. 5, we show three sequences of thunderstorm images. The first row is the source image sequence we choose, where the first image is the current source image (see Fig. 1a) and the next two images are after 10 minutes and 40 minutes, respectively. The first image in the second row is the image selected by W_1 in Section VI.A, and the following two images are its follow-up images after 10 and 40 minutes. Similarly, the first image in the third row is the image selected by W_2 in Section VI.A and the following two images are its follow-up images after 10 and 40 minutes.

Observing the figures, we notice that the sequence selected by W_1 (second row) more closely resembles the source

sequence than the one selected by W_2 (third row), especially at the 40 minute mark. However, one may also notice that the W_1 selected sequence has a noticeable shift relative to the corresponding source images. This motivates us to further study shift-invariant Wasserstein metrics in the future.

VII. Concluding Remarks and Future Work

In this paper, we thoroughly compared Wasserstein distances with some existing similarity metrics, including SSIM, FSIM, MSE, and KL. Furthermore, we focused on analyzing the 1-Wasserstein distance and applied it to identifying similar thunderstorm images. Using real radar data, we show that the 1-Wasserstein distance metric outperforms the 2-Wasserstein distance metric by matching better with human perception.

Our future work on this research includes but is not limited to: (i) Modifying the Wasserstein distance metric by removing shift and scaling effects in the image; (ii) Extend the Wasserstein-based similarity metric to sequences; and (iii) Taking into account more detailed airspace information, such as metering fixes, airport runway configuration, standard arrival and departure routes (STAR/SID) into the modeling process.

References

- [1] Dowell, D. C., Alexander, C. R., James, E. P., Weygandt, S. S., Benjamin, S. G., Manikin, G. S., Blake, B. T., Brown, J. M., Olson, J. B., Hu, M., Smirnova, T. G., Ladwig, T., Kenyon, J. S., Ahmadov, R., Turner, D. D., Duda, J. D., and Alcott, T. I., “The high-resolution rapid refresh (HRRR): An hourly updating convection-allowing forecast model. Part I: Motivation and System Description,” *Weather and Forecasting*, Vol. 37, No. 8, 2022, pp. 1371–1395.
- [2] Wang, B., Pinto, J., and Wei, P., “Identifying Similar Thunderstorm Sequences for Airline Decision Support Using Optimal Transport Theory,” *AIAA SciTech*, Vol. 1, AIAA, Orlando, FL, 2024.
- [3] Dixon, M., and Wiener, G., “TITAN: Thunderstorm identification, tracking, analysis, and nowcasting—A radar-based methodology,” *Journal of atmospheric and oceanic technology*, Vol. 10, No. 6, 1993, pp. 785–797.
- [4] Ayhan, S., and Samet, H., “Time series clustering of weather observations in predicting climb phase of aircraft trajectories,” *Proceedings of the 9th ACM SIGSPATIAL International Workshop on Computational Transportation Science*, Association for Computing Machinery, New York, NY, USA, 2016, p. 25–30. <https://doi.org/10.1145/3003965.3003968>, URL <https://doi.org/10.1145/3003965.3003968>.
- [5] Wang, P., Shi, J., Hou, J., and Hu, Y., “The Identification of Hail Storms in the Early Stage Using Time Series Analysis,” *Journal of Geophysical Research: Atmospheres*, Vol. 123, No. 2, 2018, pp. 929–947.
- [6] Hosseini-poor Milaghardan, A., Ali Abbaspour, R., and Claramunt, C., “A Spatio-Temporal Entropy-based Framework for the Detection of Trajectories Similarity,” *Entropy*, Vol. 20, No. 7, 2018. <https://doi.org/10.3390/e20070490>, URL <https://www.mdpi.com/1099-4300/20/7/490>.
- [7] Rubner, Y., Tomasi, C., and Guibas, L., “The earth mover’s distance as a metric for image retrieval,” *Int. J. Comput. Vis.*, Vol. 40, 2000, pp. 99–121.
- [8] Cuturi, M., “Sinkhorn Distances: Lightspeed Computation of Optimal Transport,” *Advances in Neural Information Processing Systems*, Vol. 26, edited by C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, Curran Associates, Inc., 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/file/af21d0c97db2e27e13572cbf59eb343d-Paper.pdf.
- [9] Chizat, L., Roussillon, P., Léger, F., Vialard, F.-X., and Peyré, G., “Faster Wasserstein Distance Estimation with the Sinkhorn Divergence,” , 2020.
- [10] Laboratory, E. O., “TORUS2022: MRMS Merged Base Reflectivity Quality Controlled Data,” <https://data.eol.ucar.edu/dataset/613.006>, 2022.
- [11] Thorpe, M., “Introduction to Optimal Transport,” https://www.damtp.cam.ac.uk/research/cia/files/teaching/Optimal_Transport_Notes.pdf, 2018.
- [12] Panaretos, V. M., and Zemel, Y., “Statistical Aspects of Wasserstein Distances,” *Annual Review of Statistics and Its Application*, Vol. 6, 2019, pp. 405–431.
- [13] Sara, U., Akter, M., and Uddin, M. S., “Image Quality Assessment through FSIM, SSIM, MSE and PSNR—A Comparative Study,” *Journal of Computer and Communications*, Vol. 7, 2019, pp. 8–18. <https://doi.org/10.4236/jcc.2019.73002>.

- [14] Markatou, M., Karlis, D., and Ding, Y., “Distance-Based Statistical Inference,” *Annual Review of Statistics and Its Application*, Vol. 8, 2021, pp. 301–327.
- [15] Snow, M., et al., “Monge’s optimal transport distance for image classification,” *arXiv preprint arXiv:1612.00181*, 2016.
- [16] Candes, E. J., Wakin, M. B., and Boyd, S. P., “Enhancing sparsity by reweighted ℓ_1 minimization,” *Journal of Fourier analysis and applications*, Vol. 14, 2008, pp. 877–905.
- [17] Greene, D. R., and Clark, R. A., “Vertically integrated liquid water—A new analysis tool,” *Monthly Weather Review*, Vol. 100, No. 7, 1972, pp. 548–552.
- [18] Amburn, S. A., and Wolf, P. L., “VIL density as a hail indicator,” *Weather and forecasting*, Vol. 12, No. 3, 1997, pp. 473–478.
- [19] Ozair, S., Lynch, C., Bengio, Y., van den Oord, A., Levine, S., and Sermanet, P., “Wasserstein Dependency Measure for Representation Learning,” *33rd Conference on Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada, 2019, pp. 15604 – 15614.